

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO5 ASSESSMENT TOOL 1 – Research Paper Review / Literature Survey

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO5	Assessment:	Assessment Tool 1 – Research Paper Review / Literature Survey
Weightage:	20%	Total Marks:	25
CO5 Statement:	Evaluate emerging technologies and societal impacts, maintaining and evolving software systems responsibly		
Student Name:	Tharankeswaran M	Reg. No.:	192424081
Date:	1/9/26	Duration:	60 minutes

Code of Conduct

I Tharankeswaran M 192424081 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Tharankeswaran M

Annexure A – Research Paper Review / Literature Survey

Instructions to Students:

Each student is assigned an individual problem statement. Based on the assigned problem statement, **conduct a literature survey and review relevant research papers** to understand the existing approaches, techniques, technologies, limitations, and research gaps related to the problem. The literature survey should include:

1. Identify and select **relevant research papers** related to the assigned problem statement from reputed journals, conferences, or scholarly databases.
2. Summarize the **objectives, methodologies, technologies, datasets, and major findings** of the selected research papers.
3. Compare the existing approaches based on suitable parameters such as **accuracy, performance, scalability, efficiency, methodology, or other problem-specific metrics**.
4. Identify the **limitations and challenges** of the existing research approaches.
5. Analyze the literature to identify the **research gap** or unresolved issues related to the assigned problem.
6. Discuss the **possible improvements or future research directions** based on the identified research gap.
7. Prepare a **comparative literature survey table** containing details such as author/year, research problem, methodology, dataset/tools, key findings, limitations, and research gap.
8. Provide proper **citations and references** using a consistent referencing style.

Deliverable: Submit a **Research Paper Review / Literature Survey Report** containing the selected research papers, individual paper reviews, comparative analysis, identified research gaps, future directions, and properly formatted references.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Selection & Review of Research Papers(25%)	Selects highly relevant, recent, credible papers and provides comprehensive reviews of objectives, methodology, tools, datasets, and findings	Selects relevant papers and provides good reviews with minor gaps	Selects moderately relevant papers with basic reviews	Papers are irrelevant or reviews are incomplete
Comparative Analysis(20%)	Provides a detailed comparison of methodologies, tools, datasets, results, and performance using a clear comparison table	Provides a good comparison with relevant parameters	Provides limited comparison with few parameters	Little or no meaningful comparison
Critical Analysis & Research Gap(25%)	Clearly analyzes limitations and identifies significant, well-supported research gaps	Identifies relevant limitations and research gaps	Identifies basic gaps with limited analysis	Research gaps are unclear or missing
Future Directions & Relevance(15%)	Proposes practical and innovative future directions directly relevant to the assigned problem statement	Proposes relevant future directions	Provides general future suggestions	Future directions are missing or unrelated
Documentation , Citations & References(15%)	Excellent academic structure, clear presentation, accurate citations, and consistent referencing	Well-organized with minor formatting/citation issues	Adequately documented with some errors	Poorly organized with inadequate or incorrect references

Criteria	Mark
Selection & Review of Research Papers(25%)	/5
Comparative Analysis(20%)	/5
Critical Analysis & Research Gap(25%)	/5

Future Directions & Relevance(15%)	/5
Documentation, Citations & References(15%)	/5
TOTAL	/25

1 Introduction and Scope of the Survey

The Smart Sports Stadium Operations Platform aims to monitor crowd movement, optimize entry and exit flows, coordinate security operations, manage parking availability, track concession performance, and generate event analytics. This literature survey reviews recent scholarly work on intelligent stadium and large-venue operations, focusing on IoT, edge computing, computer vision, multi-source data fusion, and smart parking systems. The goal is to identify existing approaches, compare their strengths and limitations, highlight research gaps, and outline future directions relevant to the assigned problem.

2 Selection of Research Papers

Six highly relevant and recent papers were selected from reputable sources (MDPI, IEEE Access, ACM, ISPRS, and IEEE conferences). Selection criteria included topical relevance to stadium/venue operations, technical depth, recency (primarily 2024–2026), and coverage of crowd monitoring, parking, multi-source fusion, or edge architectures.

1. Pattarawetwong, N., Savithi, C., & Suttidee, A. (2026). Toward an Integrated IoT–Edge Computing Framework for Smart Stadium Development. *Journal of Sensor and Actuator Networks*, 15(1), 15. <https://doi.org/10.3390/jsan15010015>
2. Alafif, T., et al. (2025). Toward an Integrated Intelligent Framework for Crowd Control and Management (IICCM). *IEEE Access*, 13, 58559–58575.
3. Zhou, S.-A. (2024). Multi-source data fusion analysis model for intelligent operation of stadiums. In *Proceedings of the 2024 International Conference on Sports Technology and Performance Analysis* (pp. 346–352). ACM. <https://doi.org/10.1145/3723936.3723989>
4. Kalbhor, A., et al. (2024). PARKTag: An AI–Blockchain Integrated Solution for an Efficient, Trusted, and Scalable Parking Management System. *Technologies*, 12(9), 155. <https://doi.org/10.3390/technologies12090155>
5. Rodríguez González, K. D., et al. (2026). UAV Remote Sensing and Deep Learning for Automatic Car Detection and Parking Occupancy Analysis: The Case of UANL Stadium, Mexico. *ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, X-3/W3-2025, 85–92.
6. Research on Computer Vision System for Intelligent Management of Football Stadium Based on Internet of Things (2024). *2024 IEEE 2nd International Conference on Sensors, Electronics and Computer Engineering (ICSECE)*.

3 Individual Paper Reviews

3.1 Paper 1: Pattarawetwong et al. (2026) – IoT–Edge Framework for Smart Stadiums

Objective: Design and evaluate a hybrid edge–cloud architecture for real-time monitoring in large national stadiums with high crowd density and legacy infrastructure constraints.

Methodology: Simulation based on the physical layout of Rajamangala National Stadium (Thailand). Integrates Automatic Number Plate Recognition (ANPR), face-recognition cameras, and panoramic cameras with edge processing nodes.

Key Technologies / Tools: IoT sensors, edge computing, cloud coordination, geometry-informed sensor placement.

Major Findings: Geometry-informed deployment minimizes blind zones without increasing camera density. Moving selected video processing tasks to the edge reduces uplink bandwidth by approximately 50–75% and stabilizes transmission under peak load. Processing location is a primary architectural design factor.

Limitations: Simulation-based evaluation; limited multi-stadium validation; facial recognition raises privacy considerations.

3.2 Paper 2: Alafif et al. (2025) – IICCM Framework

Objective: Propose an Integrated Intelligent Crowd Control and Management (IICCM) framework combining Computer Vision (CV), AI, and IoT for large-scale gatherings (tested conceptually on Hajj-scale events, transferable to stadiums).

Methodology: Four-component architecture – Crowd Vision, Crowd e-Health & IoT, Crowd Evacuation, and Cloud-IoT reliability. Emphasizes real-time identification, behaviour analysis, environmental sensing, and evacuation route modelling.

Key Technologies: CV for tracking and density estimation, AI for risk anticipation, IoT for environmental data, digital modelling of escape routes.

Major Findings: The integrated approach improves safety through proactive risk detection and optimised flow. Provides a structured blueprint for combining sensing, prediction, and intervention.

Limitations: Primarily framework-oriented; limited large-scale quantitative stadium-specific validation reported in the surveyed version; computational overhead of full CV+AI pipeline under extreme density needs further study.

3.3 Paper 3: Zhou (2024) – Multi-source Data Fusion for Stadium Operations

Objective: Overcome data dispersion and information silos in traditional stadium operations by constructing a multi-source data fusion analysis model.

Methodology: Integrates crowd flow, equipment, energy consumption, and transaction data. Employs deep-learning heterogeneous fusion, LSTM-based prediction, XGBoost/LightGBM for anomaly detection, and improved YOLOv5 for security events. Uses Redis, RabbitMQ, Spark, and ClickHouse for high-concurrency and analytics.

Key Findings: Venue utilisation increased by 15%, operating costs reduced by 12%, user satisfaction rose by 18%. Passenger-flow prediction accuracy reached 92.3%; security early-warning accuracy 96.5% with response time <100 ms.

Limitations: Empirical results are venue-specific; generalisation across different stadium sizes and event types requires additional validation; concession-specific analytics depth is limited.

3.4 Paper 4: Kalbhor et al. (2024) – PARKTag

Objective: Deliver a non-human-mediated, trusted, and scalable smart parking solution by integrating computer vision with blockchain.

Methodology: Deep-learning free-slot detection (fine-tuned models), unique vehicle identity (PARKTag), routing module, and Solidity smart contracts on Ethereum test network for automated fee calculation and ticket generation. Evaluated on a university campus dataset (>2000 images).

Key Findings: Free-slot detection validation accuracy of 92.9%. Throughput of approximately 10 user requests per second under peak load. High transparency and reduced mediation.

Limitations: Campus-scale evaluation; blockchain latency and gas costs may become constraints at true stadium scale; real-time multi-lot coordination under simultaneous ingress/egress peaks needs further testing.

3.5 Paper 5: Rodríguez González et al. (2026) – UAV + Deep Learning Parking at UANL Stadium

Objective: Automatically quantify parked vehicles and estimate occupancy using UAV imagery and deep learning at a real football stadium.

Methodology: DJI Mavic 2 and Matrice 350 RTK UAVs capture high-resolution imagery; orthomosaics (2.4–2.8 cm GSD) processed with a pretrained Mask R-CNN model. Kernel density estimation and spatial metrics generate heat maps and bottleneck indicators.

Key Findings: 4,591 vehicles detected across surveyed lots. Clear differentiation between match-day saturation and off-event occupancy. Demonstrates practical demand-responsive parking control and long-term planning support.

Limitations: UAV flights are constrained by weather, airspace regulations, and battery life; not a continuous real-time indoor/outdoor solution; requires periodic surveys rather than permanent sensor streams.

3.6 Paper 6: IEEE ICSECE 2024 – Computer Vision + IoT for Football Stadium Management

Objective: Build an intelligent management system using computer vision and IoT for comprehensive monitoring, behaviour analysis, and security early warning inside and outside football stadiums.

Methodology: Network of HD cameras, edge computing nodes for initial processing, and a central server with advanced image-recognition algorithms for density distribution, abnormal behaviour (trespassing, gathering), ticket management, and energy monitoring.

Key Findings: Significant improvement in management efficiency and safety during large-scale events; effective prevention and response to emergencies; decision support for resource allocation.

Limitations: Limited public quantitative metrics on accuracy and latency; privacy and data-protection implications of continuous visual monitoring; integration complexity with existing legacy systems.

4 Comparative Analysis

Author / Year	Core Problem	Methodology	Dataset / Tools	Key Findings	Main Limitation
Pattarawetwong et al., 2026	Real-time stadium monitoring under bandwidth & legacy constraints	Hybrid IoT–Edge–Cloud + geometry-informed sensing	Rajamangala Stadium simulation; ANPR, face & panoramic cameras	50–75% bandwidth reduction; processing location critical	Simulation-based; limited multi-site validation
Alaiff et al., 2025	Large-scale crowd safety & control	IICCM: CV + AI + IoT + evacuation modelling	Conceptual / Hajj-inspired framework	Structured proactive sensing–prediction–intervention loop	Limited quantitative stadium-scale results
Zhou, 2024	Data silos & inefficient stadium operations	Multi-source deep-learning fusion + LSTM / YOLO / Spark	Crowd, equipment, energy, transaction data	+15% utilisation, –12% cost, 96.5% security alert accuracy	Venue-specific empirical results
Kalbhor et al., 2024	Trusted, automated parking management	CV free-slot detection + Blockchain smart contracts	>2000 campus images; Ethereum testnet	92.9% slot detection accuracy; transparent ticketing	Campus scale; blockchain latency at stadium volume
Rodríguez et al., 2026	Parking occupancy at real stadium	UAV orthomosaics + Mask R-CNN	UANL Stadium lots; 4591 vehicles detected	Clear match-day vs off-event occupancy patterns	Not continuous real-time; regulatory & weather constraints
IEEE ICSECE, 2024	Intelligent football stadium management	HD cameras + edge nodes + central CV server	Football stadium environment	Improved safety & emergency response	Sparse public quantitative metrics

Table 1: Comparative Literature Survey Table

Comparison Highlights:

- **Architecture:** Edge–cloud hybrids (Pattarawetwong) and multi-source fusion platforms (Zhou) address scalability and data silos most directly.
- **Crowd focus:** Alaiff and the IEEE CV paper emphasise density, behaviour, and evacuation; Zhou adds operational KPIs.
- **Parking:** Kalbhor (CV + blockchain) and Rodríguez (UAV + Mask R-CNN) provide

complementary real-world parking solutions but lack continuous multi-lot integration with crowd and concession data.

- **Performance:** Reported accuracies are high (92–96% range for detection and alerts), yet end-to-end latency, concurrent user scale, and cross-module consistency are less uniformly reported.

5 Critical Analysis and Research Gaps

Existing research demonstrates strong individual capabilities in edge processing, computer-vision density estimation, multi-source fusion, and smart parking. However, several significant gaps remain relative to a complete Smart Sports Stadium Operations Platform:

1. **Lack of end-to-end integrated platforms:** Most works address one or two subsystems (crowd, parking, or energy). Few provide a unified architecture that simultaneously handles crowd density, entry/exit optimisation, security coordination, parking, concession performance, and analytics under a single operational dashboard.
2. **Limited real-time multi-module coordination:** Predictive models exist for density or occupancy, but closed-loop coordination (e.g., using parking and gate data to dynamically re-route crowds and adjust concession staffing) is rarely demonstrated at stadium scale.
3. **Scalability and legacy integration:** Bandwidth savings and edge processing are proven in simulation or single venues; robust patterns for heterogeneous legacy systems across different stadium sizes are still emerging.
4. **Privacy, security, and trust:** Facial recognition and continuous video analytics raise unresolved privacy and regulatory issues. Blockchain approaches (PARKTag) improve transparency for parking but have not been extended to the full operations stack.
5. **Concession and revenue analytics depth:** While transaction data appears in fusion models, fine-grained real-time concession wait-time and inventory optimisation linked to crowd heatmaps is under-explored.
6. **Standardised evaluation benchmarks:** Different papers use different datasets, metrics, and venue scales, making direct performance comparison difficult.

6 Future Research Directions

Based on the identified gaps, the following directions are particularly relevant to the Smart Sports Stadium Operations Platform:

1. Design and evaluate a unified multi-module platform that fuses crowd, parking, security, and concession data streams into a single real-time operations dashboard with predictive recommendations.
2. Develop lightweight edge–cloud pipelines that maintain sub-second latency for critical alerts while supporting privacy-preserving techniques (on-device anonymisation, federated learning).

3. Create closed-loop optimisation algorithms that jointly optimise gate allocation, parking guidance, and concession staffing based on live density forecasts.
4. Extend blockchain or verifiable logging concepts beyond parking to security incident trails and concession inventory for greater auditability.
5. Establish open, multi-stadium benchmark datasets and standardised evaluation protocols covering accuracy, latency, bandwidth, and operational impact metrics.
6. Investigate digital-twin and simulation-augmented decision support that allows operators to test “what-if” scenarios before live events.

These directions directly address the requirements of improved spectator safety, reduced waiting times, better crowd management, enhanced event experience, and more efficient stadium operations.

7 Conclusion

The surveyed literature confirms that IoT, edge computing, computer vision, multi-source data fusion, and AI-driven parking systems form a solid technological foundation for smart stadium operations. Significant progress has been made in individual modules. However, a clear research gap exists in the design and rigorous evaluation of fully integrated, real-time, multi-module platforms that coordinate crowd, security, parking, and concession functions under realistic stadium constraints. Addressing this gap through unified architectures, privacy-aware edge processing, closed-loop optimisation, and standardised benchmarks represents a promising and practically relevant future research agenda.

References

1. Pattarawetwong, N., Savithi, C., & Suttidee, A. (2026). Toward an Integrated IoT–Edge Computing Framework for Smart Stadium Development. *Journal of Sensor and Actuator Networks*, 15(1), 15. <https://doi.org/10.3390/jsan15010015>
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3. Zhou, S.-A. (2024). Multi-source data fusion analysis model for intelligent operation of stadiums. In *Proceedings of the 2024 International Conference on Sports Technology and Performance Analysis (ICSTPA '24)* (pp. 346–352). Association for Computing Machinery. <https://doi.org/10.1145/3723936.3723989>
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5. Rodríguez González, K. D., Yépez-Rincón, F. D., Escobedo Tamez, A., & Vega-Aguilar, A. (2026). UAV Remote Sensing and Deep Learning for Automatic Car Detection and Parking Occupancy Analysis: The Case of UANL Stadium, Mexico. *ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, X-3/W3-2025, 85–92.

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